Practical SBI15 and SBI16 – Rigid body docking

The goal of this practical is to use rigid-body docking to predict the structure of a protein-protein complex from the coordinates of the unbound subunits, based on geometry complementarity and FFT-based search.

Why predicting protein-protein interactions?

Most of protein functions in biological processes involve the interaction with other biomolecules and the formation of specific protein-protein complexes. Yet obtaining structural information on these complexes is technically difficult. Computer simulations can help to predict the geometry of a protein-protein complex and to understand the mechanism of protein association.

What docking programs we will use can I use?

There are several computer tools publicly available for protein-protein docking. Here we will learn about some of the most popular rigid-body docking programs:

FTDock 2.0 (http://www.sbg.bio.ic.ac.uk/docking/ftdock.html)
ZDOCK 2.1 (https://zlab.umassmed.edu/zdock/)
PatchDock (http://bioinfo3d.cs.tau.ac.il/PatchDock/)

All these programs are very fast and represented the state-of-the-art in their respective fields. These methods are very fast and can be freely used for academic research.

FTDock and ZDOCK programs use a Fast Fourier Transform (FFT) protocol to speed up the translational sampling of two proteins, in search of orientations with optimal shape complementarity. The PatchDock on the other hand is based on finding surface complementarity between the two interacting proteins. Finally, we will use Chimera to visualize and analyse the docking solutions. Other tools such ICM or PyMol are also suitable.

1. Docking Barnase and Barstar using FTDock and ZDock

Barnase is an extracellular ribonuclease and barstar is an intracellular inhibitor of barnase. To stop barnase working inside the cell, barstar binds to barnase very tightly (high affinity) and very quickly (high association rate). The binding of barnase and barstar has been very well characterized experimentally in terms of structure, energetics and kinetics. The interacting surfaces have a good surface and electrostatics complementarity, so it's not surprising that the binding is very strong and fast. That makes them a perfect target for protein-protein docking studies.

We will use here the unbound structures of barnase (PDB code 1A2P) and barstar (PDB code 1A19). The structure of the barnase-barstar complex is known (PDB code 1BRS), so we can compare the docking results with the experimental data.Thus, this will be our ground truth, so we could compare how the docking method perform.

PDB structure codes (https://www.rcsb.org)

1A2P, chain A – Unbound barnase
1A19, chain A – Unbound barstar
1BRS, chains A, D – reference barnase/barstar complex

1.1 Prepare the input files

Firstly, download the coordinates of the unbound proteins from the protein databank, PDB codes: 1A2P and 1A19.

1.1.1 Prepare coordinates files of monomers for docking

Barnase

From www.pdb.org, download coordinates of pdb code 1A2P and save them as `1a2p.pdb`. Firstly, we need to clean this file. We need to remove the lines starting with 'ANISOU' (*anisotropic temperature factors*, which may cause ZDOCK not to work correctly). You can use the following Linux command to make a file (1A2Patom.pdb) that keeps only the lines starting with 'ATOM':

```
cat 1a2p.pdb | grep ATOM > 1A2Patom.pdb
```

Now this file will contain three symmetric copies of barnase, but we need only one of the monomers for docking. Thus, you should edit the pdb file (with a standard text editor such as *vi* or *emacs*) to keep only the first chain A (water molecules and other ions can also be removed), and we will save it as `1A2P_A.pdb`.

Barstar

Download coordinates of pdb code 1A19 and save them as `1a19.pdb`. The file will contain two monomers, and we will only use one. We will save coordinates of chain A in the file `1A19_B.pdb` (with a standard text editor, as explained before). In addition, run the following command to change the chain id to 'B', so later on you will be able to visualize the docking solutions with two different chains IDs.

```
change-pdb-chain-id.perl –pdb 1A19_B.pdb –old 'A' –new 'B'
```

Note: Check that indeed chain 'A' has changed to 'B' in the PDB file.

1.1.2. Prepare coordinates of the reference complex for docking evaluation

Download coordinates of pdb code 1BRS and save them as `1brs.pdb`. The file will contain three copies of the barnase/barstar complex (the first copy is formed by molecules A and D for barnase and barstar, respectively; the second copy are molecules B and E; and the third copy are molecules C and F).

We will create a `1BRS_AD.pdb` file with one of the copies of the complex formed by chains A and D (with a standard text editor, as explained before). We will use this file later to evaluate the docking results.

Note: Check with chimera/pymol to visualize the complex and that indeed forms a binary complex.

1.2. Running ZDOCK

1.2.1 Library and script

Copy the following two files in your working directory:

```
cp /mnt/NFS_UPF/soft/zdock/IntelP3_Linux/uniCHARMM ./
cp /mnt/NFS_UPF/soft/zdock/IntelP3_Linux/create_lig ./
```

1.2.2 Formatting atomic coordinates barnase and barstar for ZDOCK

We will create new receptor and ligand pdb files (`1BRS_r.pdb` and `1BRS_l.pdb`, respectively) with the modifications required by ZDOCK:

```
/mnt/NFS_UPF/soft/zdock/IntelP3_Linux/mark_sur 1A2P_A.pdb 1BRS_r.pdb
/mnt/NFS_UPF/soft/zdock/IntelP3_Linux/mark_sur 1A19_B.pdb 1BRS_l.pdb
```

NOTE: Disregard error messages 'Unknown type' but check that the new files have been successfully created (e.g. open with a editor). Visualize both files side by side, so you can see what is modification been made.

1.2.3 Perform the docking

Issue the following command to perform the docking

```
/mnt/NFS_UPF/soft/zdock/IntelP3_Linux/zdock -R 1BRS_r.pdb -L
                1BRS_l.pdb -o 1BRS_zdock.out
```

This step takes around 5 to 10 minutes depending on the CPU load of the computer. A successful round will create a file named `1BRS_zdock.out`. Open the file with your preferred editor and inspects its contents.

```
90   1.2
0.000000  0.000000  0.000000
1BRS_r.pdb      29.181     46.629     38.264
1BRS_l.pdb      92.222     58.045     3.191
1.832596        1.977582   0.969896   77   1    14   17.32
-2.094395       1.904882   -1.478089  81   2    14   16.82
-2.356194       2.104282   -0.813972  87   85   17   16.70
1.047198        2.108018   1.311430   81   88   17   15.96
[…]
```

1.3. Running FTDock

1.3.1 Perform the docking

We will use the same two proteins as in ZDOCK .Use the following command to perform the FTDock docking, all in a single line:

```
ftdock -static 1A2P_A.pdb -mobile 1A19_B.pdb -noelec -calculate_grid
1.2 -angle_step 12 -internal -15 -surface 1.3 -keep 3 -out
1BRS.ftdock
```

Note that FTDock is slightly slower than ZDock even though we have selected parameters the following parameters to speed up calculations: -calculate_grid 1.2 large grid size:1.2 (0.875 by default) and -noelec (no electrostatics; default is to use electrostatics to score poses).

During FTDock execution, there are two intermediate files (`scratch_scores.dat` and `scratch_parameters.dat`) that are generated and used during the course of the execution. In case an FTDock execution is interrupted for any reason, these files are needed to restart such execution (by running `ftdock -rescue`).

WARNING: if you execute two FTDock runs simultaneously from the same working directory, both executions will try to generate and use the same intermediate files, which will lead to problems.To be sure to run only one FTDock execution from the same directory at the same time, and check that no `scratch_scores.dat` and `scratch_parameters.dat` files are left in the directory before starting a new FTDock execution.

The docking round should take between 15 to 30 minutes to finish. You can monitor the progress of the docking on the terminal. A successful round will create a file named `1BRS.ftdock` Open the file with your preferred editor and inspects its contents.

```
FTDOCK data file

Global Scan

Command line controllable values
Static molecule                    :: 1A2P_A.pdb
Mobile molecule                    :: 1A19_B.pdb

Global grid size                   ::     76      (user defined calculated)
Global search angle step           ::     12      (user defined)
Global surface thickness           ::      1.30   (user defined)
Global internal deterrent value    ::     15.00   (user defined)
Electrostatics                     ::    off      (user defined)
Global keep per rotation           ::      3      (user defined)

Calculated values
Global rotations                   ::   9240
Global total span (angstroms)      ::     91.522
Global grid cell span (angstroms)  ::      1.204

Data
Type        ID    prvID    SCscore        ESratio         Coordinates            Angles

G_DATA       1        0      17838          0.000        -1   -2    0       156 108 240
G_DATA       2        0      17811          0.000        -1   -1    1       156 108 240
G_DATA       3        0      17757          0.000        -1   -1   -1        24 156 288
[…]
```

2. Analysis of ZDOCK and FTDOCK docking solutions

We will analyze here the docking solutions generated by FTDock and ZDOCK programs in previous two sections.

2.1. Create the pdb files from the ZDOCK docking output

We can create pdb files from the docking conformations in the ZDOCK output file. The ZDOCK output file contains information for 2000 docking solutions, so using the unmodified output would create 2000 docking files in your current directory. It is thus advisable to edit the ZDOCK output file to keep only the top N solutions (for instance, N=10). You can use a standard text editor for that, or the following linux command (forinstance, if you want to keep the top 10 solutions, i.e. N=10):

```
head -n 14 1BRS_zdock.out > 1BRS_zdock_10.out
```

Note that we use "-n 14" to keep also the first 4 lines (in addition to the 10 lines with the top 10 solutions), which contain important information.

Now we can create pdb files for the top 10 docking solutions from the short ZDOCK output file (pdb files will be named *complex.1*, *complex.2* and so on...):

```
/mnt/NFS_UPF/soft/zdock/IntelP3_Linux/create.pl 1BRS_zdock_10.out
```

Note: The `create_lig` and `uniCHARMM` files copied in section 1.2.1 should be in the same directory as the ZDOCK files.

NOTE: If the chains are separated by an 'END' tag, perform the following(1). Otherwise, you can just rename the files:

```
mv -v complex.1 complex.1.pdb
mv -v complex.2 complex.2.pdb
        etc … up to
mv -v complex.10 complex.10.pdb
```

(1) Change the format of the output files, so it can be read directly into Chimera (pdb files will be named *complex.1.pdb*, *complex.2.pdb* and so on...):

```
arrange.pl complex.1 complex.1.pdb
arrange.pl complex.2 complex.2.pdb
arrange.pl complex.3 complex.3.pdb
arrange.pl complex.4 complex.4.pdb
arrange.pl complex.5 complex.5.pdb
arrange.pl complex.6 complex.6.pdb
arrange.pl complex.7 complex.7.pdb
arrange.pl complex.8 complex.8.pdb
arrange.pl complex.9 complex.9.pdb
arrange.pl complex.10 complex.10.pdb
```

2.2. Create the pdb files from the FTDock docking output

We can create pdb files from the FTDock output file:

```
build -in 1BRS.ftdock -b1 1 -b2 10
```

This command will create pdb files for the top 10 docking solutions (from 1 to 10) in the FTDock output file (pdb files will be named *Complex_1g.pdb*, *Complex_2g.pdb*, etc).

2.3. Visualization of docking solutions

The objective is to compare the top 10 dockings poses of both FTDock and ZDOCK and compare with the reference complex: the solved Xray structure of the Barnase/Barstar complex you prepared earlier: 1BRS_AD.pdb.

In the previous sections you have run the docking and extracted the top 10 docking poses of both docking programs. Making use of chimera you should compare these docking poses with real complex. You will have to load the docking poses and the Xray complex in chimera* and perform the structural superposition using the Barnase as reference.

Consider the following question in your analyses:

- How close are the highest-scoring FTDock and ZDOCK solutions to the x-ray structure?

- How many docking solutions with RMSD<10Å can be found in the top 10 solutions from each program?

- Will the combination of results from both programs improve the chances of a good prediction?

- From this analysis on a test case in which complex structure is known, have we learned any suggestion to optimally apply these docking programs in a real case situation?

- Identify the closest docking pose among the top 10 solution from each program to the real complex.

> **Note.** You can also use Pymol to structurally align and compute RMSD values to analyze your docking solutions. In the campus, you can find a quick 'how-to' to achieve this with pymol (quick how to structural superposition pymol.pdf)

3. Docking of the ovine prion protein and a neutralizing antibody using PathDock

Now we will use *PatchDock*, a useful and popular rigid-body docking program that is based on the identification of surface complementarity between the two interacting proteins. We will perform the docking of an antibody to the ovine prion.

Prion diseases are deadly neurodegenerative pathologies affecting numerous mammal species [1]. The key event in the pathogenesis is the conversion of the α-helix-rich host prion protein (PrPc) into a pathogenic isoform (PrPSc) characterized by its insolubility, its high content in β-sheet, and its protease resistance.

3.1. Input files for docking

Download the `SBI_15_P.zip` from the Aula global. It contains the following files:

- antibody.pdb - atomic coordinates of the antibody
- ovine.pdb – atomic coordinates of the ovine prion: antigen
- NativeComplex.pdb – atomic coordinates of the Xray complex (1TPX)

3.2 Running PatchDock

<u>3.2.1 Create the parameter file that is required by PatchDock</u>

```
buildParams.pl antibody.pdb ovine.pdb 4.0 AA
```

This step creates a text file: `params.txt` with all the parameters required to run PatchDock. First and second arguments are the input atomic coordinates in the form of receptor and ligand, RMSD cut-off clustering and type of complex. In this case is `'AA'` which stands for Antigen-Antibody complex. There are other types considered by PatchDock: EI: Enzyme/Inhibitor; drug: protein/ligand; default: protein/protein (neither AA nor EI).

<u>3.2.2. Perform the docking</u>

Issue the following command in a single line

```
patch_dock.Linux params.txt out_file1
```

PathDock is very fast and the docking will be done in a few minutes. The results will be store in the file: `out_file1`

Open the file with a text editor and analyze the format of the file:

```
*******************************************************
Program parameters

ACE_EnergyTermWeight     (Str)   1.0
COM_distanceTermWeight   (Str)   1.07
HBEnergyTermWeight       (Str)   1.0
attrVdWEnergyTermWeight  (Str)   1.01
baseParams       (Str)   4.0 13.0 2
clusterParams    (Str)   0.1 4 2.0 4.0
confProbEnergyTermWeight         (Str)   0.1
desolvationParams        (Str)   500.0 1.0
```

```
elecEnergyTermWeight    (Str)   0.1
energyDistCutoff        (Str)   6.0
ligandActiveSite        (Str)   site2.txt
ligandGrid      (Str)   0.5 6.0 6.0
ligandMs        (Str)   carly1.pdb.ms
ligandPdb       (Str)   carly1.pdb
ligandSeg       (Str)   10.0 20.0 1.5 1 1 1 1
log-file        (Str)   patch_dock.log
log-level       (Str)   2
matchAlgorithm  (Str)   1
matchingParams  (Str)   1.5 1.5 0.4 0.5 0.9
piStackEnergyTermWeight (Str)   0.0
protLib (Str)   /home/narcis/DOCKINGS_CD19/PatchDock/chem.lib
radiiScaling    (Str)   0.8
receptorActiveSite      (Str)   site1.txt
receptorGrid    (Str)   0.5 6.0 6.0
receptorMs      (Str)   CD19.pdb.ms
receptorPdb     (Str)   CD19.pdb
receptorSeg     (Str)   10.0 20.0 1.5 1 1 1 2
repVdWEnergyTermWeight  (Str)   0.5
scoreParams     (Str)   0.3 -5.0 0.5 0.0 0.0 1500 -8 -4 0 1 0
vdWTermType     (Str)   1

*********************************************************


  # | score | pen.  | Area    | as1   | as2   | as12  | ACE     | hydroph | Energy  |cluster| dist. || Ligand Transformation
  1 | 11502 | -3.26 | 1623.00 |  1797 |  2106 |   683 |  313.12 | 1146.31 |    0.00 |     0 | 0.00 || -0.05752 1.19704 -2.20131 -37.29977 -2.48074 -3.56842
  2 | 11290 | -3.78 | 1525.60 |  2331 |  1821 |  1333 |  191.56 | 1013.13 |    0.00 |     0 | 0.00 || 0.04000 0.12658 -2.77974 -34.22400 -16.34312 3.45601

[…]
```

Notes: PatchDock is also accessible through a dedicated server:

http://bioinfo3d.cs.tau.ac.il/PatchDock/php.php

3.2.3 Generate the top 10 docking solutions

To generate the top 10 docking solutions simply issue the following command in a single line:

```
transOutput.pl out_file1 1 10
```

3.3 Visualization and analysis of the PatchDock results

As in the previous exercise open the top 10 docking solutions and the reference complex. Superimpose all the structures and analyze the results. Are the results reasonable/accurate? Which docking poses among the top 10 is the closest to the reference complex?

REFERENCES

[1] Prusiner, S. B. (1998) *Proc. Natl. Acad. Sci. USA* 95:13363-13383.
[2] Janin J. (2005) *Proteins*. 1;60(2):170-5.
[3] Eghiaian et al. (2004) *Proc Natl. Acad. Sci. USA.* 101(28):10254-9.

COMPLEMENTARY WORK – Predicting binary complexes with AF3

As you probably know, recent years have witnessed a major revolution with the development of AI-assisted technologies to predict the 3D structures of proteins and protein complexes. In particular, AlphaFold[1] and subsequent versions: AlphaFold2[2] and AlphaFold3[3] has been established as the most successful and accurate methodology in the field.

Due to technical reasons and computational limitations, we cannot perform the prediction of structural complexes in a course like this one. However, it is possible to run Alphafold3 using the public available server provided by DeepMind: https://alphafoldserver.com/

To use the Alphafold3 server you only need a Gmail account that allow the user to submit up to 20 jobs. The interface is very intuitive; you just need to provide the sequences of the proteins that form the complex and the server return the top scoring predictions.

You could try to predict both the Barnase/Barstar and Ovine prion – Antibody complexes and compare to the models you have obtained using the traditional docking approaches as well as the real complexes.

Is AF3 able to predict the native complex with high accuracy and reliability?

Is there any difference between the Barnase/Barstar and Ovine prion – Antibody?

Is AF3 better predicting one of the other? Any particular reason(s)?

What conclusions can you extract from these analyses?

[1] Senior et al. (2020) *Nature* 577(7792):706-710
[2] Jumper el al. (2021) *Nature* 596(7873):583-589
[3] Abramson et al. (2024) *Nature* 630(8016):493-500